

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've been writing software for about 10 years. The last three I've spent as a Software Architect owning backend services and REST APIs in TypeScript and Python on AWS, at a company with over \$300M in annual transactions. I own 4 APIs end to end: design, OpenAPI contracts, versioning, and the part after launch. Testing has been part of how I work since early in my career. I designed a test framework that reduced defects by 30% and is still running on the team today, 4+ years later. I build internal tooling from scratch when existing solutions don't fit, use AI coding tools every day, and I care about shipping things end to end from design doc to deployment.

Work Experience

Alberta Motor Association

SOFTWARE ARCHITECT

Edmonton, Alberta

Jan 2023 - Present

- Own 4 REST APIs serving 5 internal teams and external partners. Designed the contracts and versioning, approve new endpoint additions, and maintain OpenAPI design standards in Confluence. (Python, REST, OpenAPI, TypeScript)
- Built a self-service developer platform solo that took environment provisioning from 2-3 weeks down to 2 hours (60x). Self-service create/destroy, database snapshots, and pull-from-prod, adopted by 15 people across 3 teams and used to spin up 120+ environments. It made production debugging routine: a cancellation bug nobody could reproduce got root-caused in days using snapshotted production data. (Python, Terraform, AWS, Jenkins, Docker)
- Led a 2-3 engineer rewrite of a brittle legacy C# document generation system into containerized .NET microservices. 4-5 services (with sidecars) on a managed Kubernetes cluster, deployed with Helm. Wired up Datadog metrics for failure rate, throughput, generation time, and storage time, replacing a system that had frequent outages. (C#/.NET, Kubernetes, Docker, Helm, Datadog)
- Set technical standards for a 12-person team through code reviews, RFC authorship, and PR checklists / API design guides. Authored an RFC moving the team to trunk-based development with feature flags. Worked in cross-functional pods alongside PM, design, and QA. Member of the Architecture Steering Committee, and mentor newer developers through legacy codebases and TDD. (Gosu/Java)
- Architected the infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, 99.9% uptime since launch, 2.8M insured trips and 76,637 invoices processed. Won the 2024 Guidewire Innovation Award (1 of 3 worldwide). (Python, AWS Lambda, Terraform, TypeScript)
- Built an event-driven pipeline that ingests an external partner's policy data: S3 ingestion, a Kubernetes CronJob, a translation service, and transactional calls to a composite API, with OAuth 2.0 auth and Datadog monitoring. (Python, Kubernetes, AWS S3, OAuth 2.0)

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built and maintained backend services that ran medical AI algorithms and returned results to the platform. (TypeScript, Node.js, AWS)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework that reduced defects by 30% on subsequent projects. Adopted across the insurance engineering team (15 developers / all Guidewire projects) and still in use 4+ years later. (Gosu/Java)
- Led 5-6 person development team on Guidewire insurance platform using TDD methodology. Performed Scrum Master role. Delivered projects on time and budget with no critical defects post-launch. Code reviews and mentoring junior developers.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local Guidewire development environments. Trained 5-6 developers on TDD methodologies through hosted training sessions.

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET, heavily utilizing the MVC design pattern, with REST integration and tests written with NUnit and Moq. (C#, .NET, MVC, REST)

Lockheed Martin Canada

ASSOCIATE SOFTWARE ENGINEER

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above-water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging and testing capabilities.

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

Education

The Open University MSC. COMPUTING (SOFTWARE ENGINEERING) Currently pursuing Master's in Computing at The Open University.	Milton Keynes, England (Remote) May 2024 - Ongoing
The King's University B.S. IN COMPUTER SCIENCE	Edmonton, Alberta Sep 2012 - Dec 2016

Skills

Backend Services & Ownership	I've spent the last three years owning REST APIs and backend services in TypeScript and Python end to end: design, OpenAPI contracts, versioning, integration points, and the part after launch. I led a 2-3 engineer rewrite of a brittle legacy document generation system into containerized microservices. I like building internal tools from scratch when existing solutions don't fit, and I care about leaving systems more maintainable than I found them.
Testing Culture	TDD has been part of how I work since early in my career. I designed and built a test framework that reduced defects by 30% on subsequent projects and is still in use today, 4+ years later. I write unit, integration, and end-to-end tests (Playwright). Testing culture is something you have to model, not just advocate for, and I've run training sessions on TDD methodology for 5-6 developers.
AI-Forward Development	I use AI coding assistants and agentic workflows every day, and I've formed real opinions on where they speed things up and where they quietly create work. I'm comfortable folding them into the development lifecycle and helping a team get good at using them well, with guardrails rather than blind trust.
Technical Leadership & Design Docs	I write the design docs and RFCs that align a pod on a hard call, including an RFC to move my team to trunk-based development with feature flags so we could decouple deploys from releases. I set standards through code reviews, PR checklists, and API design guides, and I sit on AMA's Architecture Steering Committee. Making the engineers around me better through clear writing and useful reviews is a real part of how I work.

Technologies

Backend Languages	TypeScript, Python, Node.js, C# / .NET, Java / Gosu (JVM), Bash
Architecture & APIs	RESTful APIs, OpenAPI, Microservices, Event-Driven Architecture, API Versioning
Testing	TDD, Playwright (E2E), Unit Testing, Integration Testing, Test Framework Design
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM, ECR), Terraform, Kubernetes, Docker, Azure DevOps, Azure Artifacts
Databases	PostgreSQL, SQL, Aurora/RDS, Data Modelling, Redis
Observability	Datadog, Elasticsearch, Kibana, Metrics Design, SLOs, Alerting
CI/CD & Automation	CI/CD, GitHub Actions, Jenkins, GitOps, Feature Flags, Infrastructure as Code
AI-Assisted Development	Claude Code, Cursor, GitHub Copilot, Agentic Workflows